Problem-Solving Tutorial 2

Uncertainty Relations and Orbital Angular Momentum

The problem-solving tutorials will have a mix of group discussion questions and questions for individual (or small group study). You are welcome to work with your friends and neighbours, and please ask the tutor questions. We want these sessions to be as interactive as possible.

1. **[Discussion in Tutorial]** Please look at the statements, and identify why they are wrong or "not quite right":
   (a) We have some matrix $H_{nm} = \langle n|\hat{H}|n\rangle$, the first element would be $H_{11} = \langle 1|\hat{H}|1\rangle$.
   (b) Let $\hat{O} = \langle[\hat{x}, \hat{p}]\rangle$.
   (c) $(\hat{a}_- + \hat{a}_+)^2 = \hat{a}_-^2 + \hat{a}_+^2 + 2\hat{a}_-\hat{a}_+$

2. **[Individual - at home, check in tutorial]** In this question, we will look at finding the uncertainty for the momentum operator for a harmonic oscillator in the state:
$$|\psi\rangle = \frac{1}{\sqrt{6}}|0\rangle + \frac{2}{\sqrt{6}}|1\rangle + \frac{1}{\sqrt{6}}|2\rangle$$
   Hint! You may want to look at assessed problem sheet 2.
   (a) Calculate $< \hat{p} >$.
   (b) Then, calculate $< \hat{p}^2 >$.
   (c) Finally, calculate $\Delta\hat{p}$.

3. **[Individual - exam style question for tutorial]** In week 5 we looked at the specific case of Orbital Angular Momentum (OAM) before we delved into the generalised picture. Here, I would like you to work through some key commutator relations of OAM:
   (a) Show $[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z$. Is this apparent cyclic relationship true for every combination of x, y, z? What would we expect to see for $[\hat{L}_y, \hat{L}_x]$, for instance?

(b) Show that for any component of $\hat{\mathbf{L}}$ that, they do indeed commute with $\hat{\mathbf{L}}^2$.

(c) Finally, show $[\hat{L}^2, \hat{H}] = [\hat{L}_z, \hat{H}] = 0$. Where $\hat{H}$ is the Hamiltonian for a moving particle in a given potential $V(r)$:

$$\hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r}).$$

Hint! Rewriting both $\hat{L}^2, \hat{L}_z$ and $\hat{H}$ in polar co-ordinates is a good place to start.